

Roberto Segura Sala

DATA PLATFORMS / APPLIED AI / TECHNICAL LEADERSHIP

Madrid, Spain · Engineering since 2011

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I combine platform-wide technical direction with hands-on engineering and collaboration with product teams. I enjoy continuous learning, complex problems, and turning loosely defined needs into concrete solutions with a high degree of autonomy.

Professional experience

Audiense | Backend, Big Data & Real Time Engineer

Jan 2021 - Present

Technical leader for the data platform, combining architecture, implementation, cross-team adoption, and mentoring.

- Led the adoption of Trino and Apache Iceberg, creating and maintaining terabyte-scale tables and coordinating dozens of real-time consumers. Completed the migration to real-time enrichment.
- Drove VEGA adoption by product teams, reducing costs and reliance on separate systems while enabling new data-driven product verticals previously constrained by scale.
- Improved query performance through table statistics and join optimization, lowering network traffic and CPU use; introduced retries to reduce query failures.
- Expanded interest enrichment to more profiles and improved assignment correctness through LoRA fine-tuning and reranking, with model serving on Kubernetes and vLLM.
- Mentors engineers in AI-assisted development and supports teams in adopting the data platform.

TMC | Data Platform Engineer

Sep 2020 - Jan 2021

Future Space | Data Platform Engineer

Sep 2019 - Sep 2020

Conceptualized and developed a production platform for fraud analysis in motor insurance claims using Scala, Kafka, Akka, and Cassandra.

Earlier experience

2011 - 2019

everis | Big Data Engineer | Feb 2018 - Sep 2019

Batch and streaming systems, Kafka Streams model deployment, ELK monitoring.

Finect | Senior Developer | Jan - Oct 2017

Node.js APIs, OAuth2, and data analysis planning.

playthe.net | Big Data Engineer | May 2014 - Dec 2016

Real-time football analytics with Spark Streaming and Cassandra; IoT integration.

Primum Health IT | Liferay Engineer | Aug 2013 - Apr 2014

amaranto Consultores | Senior Developer | Dec 2011 - Aug 2013

UCLM | Junior Developer | Jan - Aug 2011

Engineering, learning & exploration

SELECTED TECHNICAL WORK

Streaming data & asynchronous query architecture

Iceberg data lake maintained through streaming enrichment. VEGA unifies Athena and Trino behind an asynchronous API with queue-based decoupling, execution traceability, retries, and Redis rate limiting.

Applied machine learning & inference

Interest classification across 444 categories using LoRA-tuned embeddings, cross-encoder reranking, and model agreement. Evaluation F1 improved from 0.255 to 0.390 after fine-tuning; strict precision rose from 36% to 52% with model agreement. Kubernetes and vLLM serving connect inference to streaming consumers; Qwen models support near-real-time enrichment.

Technical expertise

Data: Kafka, Redpanda, Kafka Streams, Spark, Iceberg, Trino, Athena, Parquet.

AI: Sentence Transformers, LoRA/PEFT, reranking, spaCy, vLLM, Qwen, agent harnesses.

Backend & infrastructure: Python, Scala, SQL, Node.js, FastAPI, Akka HTTP, AWS, Kubernetes, Docker, Terraform, Redis, Cassandra, SQLite.

Leadership: architecture direction, platform adoption, product collaboration, technical mentoring.

Personal projects & research

- **Hermes Expense Tracker:** conversational household expense tracking with per-member Hermes assistants, a Python FastMCP domain server, SQLite, and Telegram/CLI access.
- **Voice Agent Architecture Prototype:** hands-on exploration of STT, LLM orchestration, TTS, and durable conversation state using LiveKit, Deepgram, and ElevenLabs.
- **Math Tutor:** voice-agent prototype with a validation harness and deterministic educational rules, separated from model providers and tested through offline evaluations.
- **AI Research Wiki:** Obsidian knowledge vault developed with Hermes Agent, linking sources, concepts, comparisons, and research syntheses.

Education & selected learning

Master in Big Data Development | Hadoop Training - UCLM | 2015 - 2016

Bachelor's Degree in Telecommunications | UCLM

Data Streaming Nanodegree - Udacity, 2021.

Confluent Certified Developer for Apache Kafka - awarded 2018; expired Jan 2022.

Generative AI and LLMs: Architecture and Data Preparation - IBM, 2026.

Recent reading

Currently reading: Domain-Specific Small Language Models.

Completed: Designing Data-Intensive Applications; Build a Large Language Model (From Scratch); Build an AI Agent (From Scratch).